

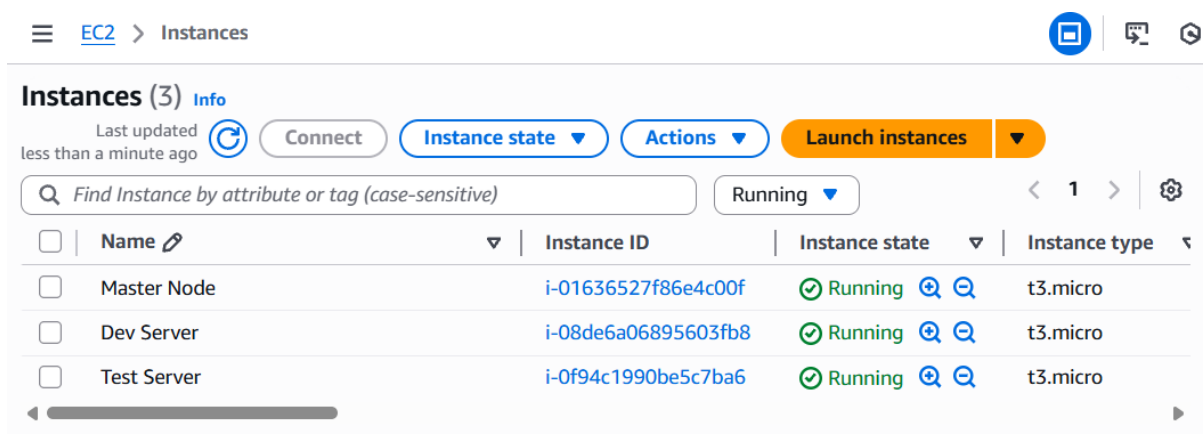
01 Assignment - Configuration

Automation using Ansible

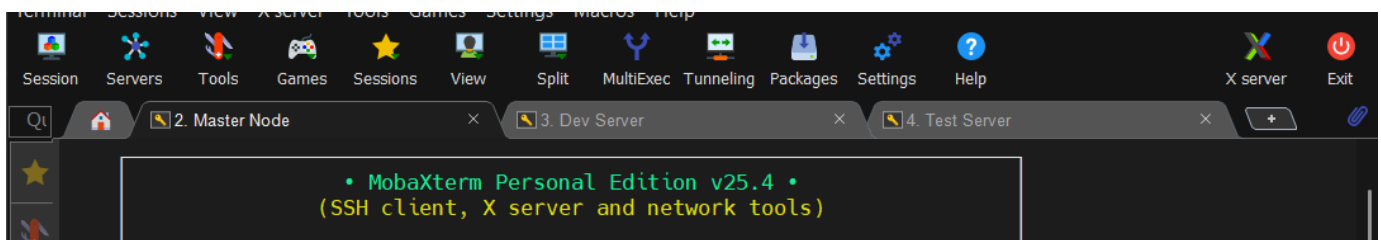
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1. L1 - Create and Execute Ansible Playbook to Setup Java Maven Application Build Server

- Launch 3 EC2 instances: one as the **Control Node** (Master) and two as **Worker Nodes** (Dev Server and Test Server).



- Connect to all three nodes using **MobaXterm**.



- Set the hostname for all nodes and switch to root using `sudo -i`.

Multi-execution mode: commands are typed to all terminals

Multi-paste Exit multi-execution mode

```
[root@ip-172-31-5-185 ec2-user]# hostnamectl set-hostname master
[root@ip-172-31-5-185 ec2-user]# sudo -i
[root@master ~]#
```

☐ Exclude "Master Node" from MultiExec mode

```
[root@ip-172-31-6-240 ec2-user]# hostnamectl set-hostname dev-server
[root@ip-172-31-6-240 ec2-user]# sudo -i
[root@dev-server ~]#
```

☐ Exclude "Dev Server" from MultiExec mode

```
[root@ip-172-31-15-145 ec2-user]# hostnamectl set-hostname test-server
[root@ip-172-31-15-145 ec2-user]# sudo -i
[root@test-server ~]#
```

☐ Exclude "Test Server" from MultiExec mode

- Install **Ansible** on the Control Node.

```
[ec2-user@master ~]$ ansible --version
ansible [core 2.15.13]
  config file = None
  configured module search path = ['/home/ec2-user/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/local/lib/python3.9/site-packages/ansible
  ansible collection location = /home/ec2-user/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/local/bin/ansible
  python version = 3.9.25 (main, Dec 10 2025, 00:00:00) [GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] (/usr/bin/python3)
  jinja version = 3.1.6
  libyaml = True
```

- Edit the SSH configuration file `/etc/ssh/sshd_config`:
 - Set `PermitRootLogin` yes (line 38)
 - Set `PasswordAuthentication` yes (line 63)

This allows commands to be run on other servers as the root user

```
# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes
```

```
2. Master Node 3. Dev Server 4.
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# Explicitly disable PasswordAuthentication. By presetting it, we
# avoid the cloud-init set_passwords module modifying sshd_config and
# restarting sshd in the default instance launch configuration.
PasswordAuthentication yes
PermitEmptyPasswords no

# Change to no to disable s/key passwords
#KbdInteractiveAuthentication yes
```

- Apply the SSH configuration changes to **all nodes**.
- Restart the SSH service on all nodes for changes to take effect

```
2. Master Node 3. Dev Server 4. Test Server
[root@master ~]# vi /etc/ssh/sshd_config
[root@master ~]# systemctl restart sshd
[root@master ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
   Active: active (running) since Wed 2026-02-04 13:56:55 UTC; 33s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 16717 (sshd)
     Tasks: 1 (limit: 1067)
    Memory: 1.4M
       CPU: 9ms
    CGroup: /system.slice/sshd.service
            └─16717 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Feb 04 13:56:55 master systemd[1]: Starting sshd.service - OpenSSH server daemon...
Feb 04 13:56:55 master sshd[16717]: Server listening on 0.0.0.0 port 22.
Feb 04 13:56:55 master sshd[16717]: Server listening on :: port 22.
Feb 04 13:56:55 master systemd[1]: Started sshd.service - OpenSSH server daemon.
[root@master ~]#
```

- Set passwords for all nodes to enable root access.

The screenshot shows a multi-terminal window titled "Multi-execution mode: commands are typed to all terminals". It contains three terminal panes, each showing the execution of the `passwd root` command. The output for each terminal is identical: "Changing password for user root. New password: BAD PASSWORD: The password is shorter than 8 characters. Retype new password: passwd: all authentication tokens updated successfully." The panes are labeled with checkboxes to exclude specific nodes from the multi-execution mode: "Exclude 'Master Node' from MultiExec mode", "Exclude 'Dev Server' from MultiExec mode", and "Exclude 'Test Server' from MultiExec mode".

```
[root@master ~]# passwd root
Changing password for user root.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@master ~]#
```

```
[root@Dev-Server ~]# passwd root
Changing password for user root.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@Dev-Server ~]#
```

```
[root@Test-Server ~]# passwd root
Changing password for user root.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@Test-Server ~]#
```

- Generate an **SSH key** on the Control Node using `ssh-keygen`

The screenshot shows a terminal window with three tabs: "2. Master Node", "3. Dev Server", and "4. Test Server". The active tab is "2. Master Node", which shows the execution of the `ssh-keygen` command. The output indicates that a public/private RSA key pair is being generated, saved in `/root/.ssh/id_rsa` and `/root/.ssh/id_rsa.pub`. The key fingerprint is displayed as `SHA256:VlrG7mtn2MN95F0Fk+k1pMZjjAqBK8WLnVjS4tam6zs` for `root@master`. The key's randomart image is also shown.

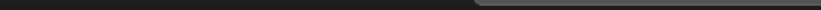
```
[root@master ~]# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa
Your public key has been saved in /root/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:VlrG7mtn2MN95F0Fk+k1pMZjjAqBK8WLnVjS4tam6zs root@master
The key's randomart image is:
+---[RSA 3072]-----+
|  o .. .+. |
| o * .. +*.. |
| . X +. =. 0 +. |
| * B .*. o o . |
| . + S.. . |
| . . . 0 |
| . .+. . oo |
| E o.* . + |
| .oo ..o . . |
+----[SHA256]-----+
[root@master ~]#
```

- ```
[root@master ~]# ssh-copy-id root@172.31.5.252
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/root/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
root@172.31.5.252's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh 'root@172.31.5.252'"
and check to make sure that only the key(s) you wanted were added.
```
- ```
[root@master ~]# ssh 172.31.5.252
```
- ```
 ##_
 _###
 ~\#####
 ~~_#####\
   ~~~\####|
   ~~~\#/
 V~' '-> https://aws.amazon.com/linux/amazon-linux-2023

 .
 / \
 / \
 /m/' \
```
- Last login: Wed Feb 4 13:17:34 2026  
[root@Dev-Server ~]# █

- 
- ```
[root@master ~]# sudo mkdir -p /etc/ansible
[root@master ~]# sudo nano /etc/ansible/ansible.cfg
```

```
2. Master Node 3. Dev Server
[defaults]
inventory=/etc/ansible/hosts
host_key_checking=False
~
~
~
~
```

```
GNU nano 8.3 /etc/a
[dev-server]
172.31.5.252

[test-server]
172.31.11.110
```

- Test the connection from Control Node to Worker Nodes using:
ansible all -m ping Expected output: {"ping": "pong"}

```
2. Master Node 3. Dev Server 4. Test Server
[root@master ~]# sudo mkdir -p /etc/ansible
[root@master ~]# sudo nano /etc/ansible/ansible.cfg
[root@master ~]# vi /etc/ansible/ansible.cfg
[root@master ~]# sudo nano /etc/ansible/hosts
[root@master ~]# ansible all -m ping
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to see details
[WARNING]: Platform linux on host 172.31.5.252 is using the discovered Python interpreter at
/usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that
path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for
more information.
172.31.5.252 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[WARNING]: Platform linux on host 172.31.11.110 is using the discovered Python interpreter at
/usr/bin/python3.9, but future installation of another Python interpreter could change the meaning of that
path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for
more information.
172.31.11.110 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.9"
  },
  "changed": false,
  "ping": "pong"
}
[root@master ~]#
```

- Create an **Ansible playbook** on the Control Node to install **Java** and **Maven** on the Worker Nodes.

```

---
- name: Setup Java Maven Build Server
  hosts: all
  become: yes

  tasks:
    - name: Install Java (OpenJDK 11)
      yum:
        name: java-11-amazon-corretto-devel
        state: present

    - name: Print Message
      debug:
        msg: "Java Installed Successfully"

    - name: Install Maven
      yum:
        name: maven
        state: present

    - name: Print Message
      debug:
        msg: "Maven Installed Successfully"
...
~
~
~
~
~
~

```

- Execute the playbook using *ansible-playbook <yaml filename>* command.

```

TASK [Install Java (OpenJDK 11)] *****
changed: [172.31.11.110]
changed: [172.31.5.252]

TASK [Print Message] *****
ok: [172.31.5.252] => {
  "msg": "Java Installed Successfully"
}
ok: [172.31.11.110] => {
  "msg": "Java Installed Successfully"
}

TASK [Install Maven] *****
changed: [172.31.5.252]
changed: [172.31.11.110]

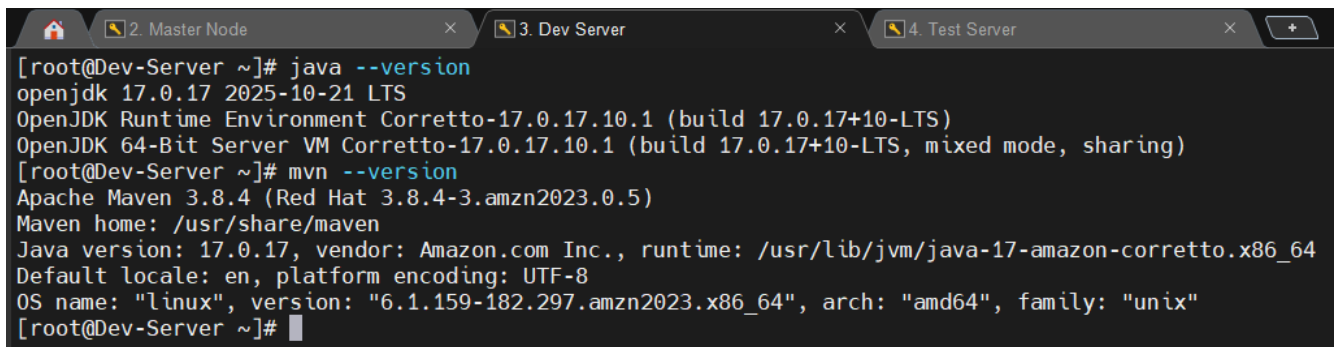
TASK [Print Message] *****
ok: [172.31.5.252] => {
  "msg": "Maven Installed Successfully"
}
ok: [172.31.11.110] => {
  "msg": "Maven Installed Successfully"
}

PLAY RECAP *****
172.31.11.110      : ok=5    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ign
ored=0
172.31.5.252      : ok=5    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ign
ored=0

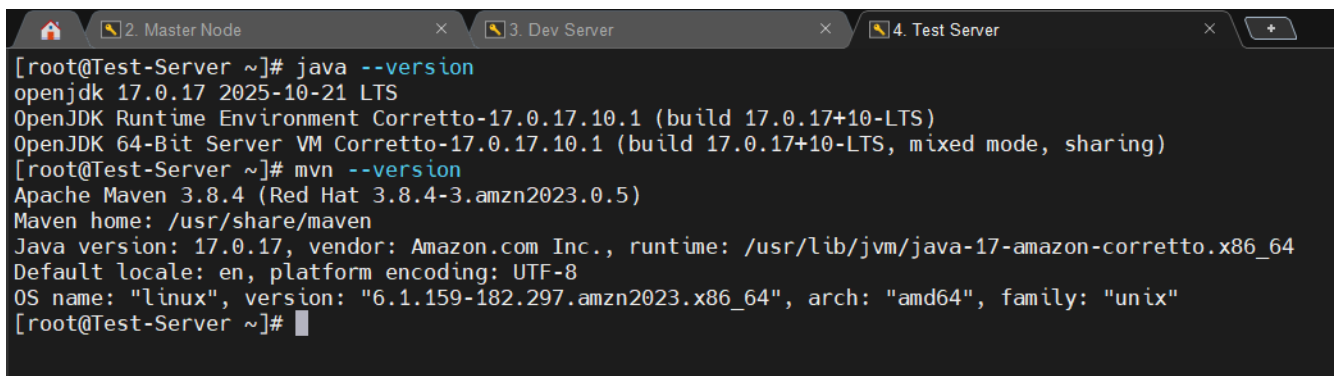
[root@master ~]#

```

- Verify the installations on the Worker Nodes.



```
[root@Dev-Server ~]# java --version
openjdk 17.0.17 2025-10-21 LTS
OpenJDK Runtime Environment Corretto-17.0.17.10.1 (build 17.0.17+10-LTS)
OpenJDK 64-Bit Server VM Corretto-17.0.17.10.1 (build 17.0.17+10-LTS, mixed mode, sharing)
[root@Dev-Server ~]# mvn --version
Apache Maven 3.8.4 (Red Hat 3.8.4-3.amzn2023.0.5)
Maven home: /usr/share/maven
Java version: 17.0.17, vendor: Amazon.com Inc., runtime: /usr/lib/jvm/java-17-amazon-corretto.x86_64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.1.159-182.297.amzn2023.x86_64", arch: "amd64", family: "unix"
[root@Dev-Server ~]#
```



```
[root@Test-Server ~]# java --version
openjdk 17.0.17 2025-10-21 LTS
OpenJDK Runtime Environment Corretto-17.0.17.10.1 (build 17.0.17+10-LTS)
OpenJDK 64-Bit Server VM Corretto-17.0.17.10.1 (build 17.0.17+10-LTS, mixed mode, sharing)
[root@Test-Server ~]# mvn --version
Apache Maven 3.8.4 (Red Hat 3.8.4-3.amzn2023.0.5)
Maven home: /usr/share/maven
Java version: 17.0.17, vendor: Amazon.com Inc., runtime: /usr/lib/jvm/java-17-amazon-corretto.x86_64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.1.159-182.297.amzn2023.x86_64", arch: "amd64", family: "unix"
[root@Test-Server ~]#
```

- This task successfully set up a Java-Maven build server environment across multiple nodes using Ansible. The Control Node was able to manage the Worker Nodes efficiently, and both Java and Maven were installed and verified.